

Design & Analysis of Algorithms

Lab 1: Algorithm Performance Analysis Workbench

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Q1. 1.4 Algorithm Execution Observation Table (5 marks)

```
from math import floor, log2
```

```
def generate_execution_observation_table(sizes):
    # your code goes here
    def rfib(n):
        if n<=1:
            return 1
        return 1+ rfib(n-1)+rfib(n-2)
    output=[
        "Algorithm Execution Observation Table",
        "InputSize RecursiveFactorial IterativeFactorial RecursiveFibonacci IterativeFibonacci
        LinearSearch BinarySearch BubbleSort InsertionSort"
    ]
    for n in sizes:
        rfact=n+1
        ifact=n
        rfibc=rfib(n)
        ifib=n
        lsearch=n
        bsearch=floor(log2(n))+1
        bubble=n*(n-1)//2
        insertion=n*(n-1)//2
        row=f"{n} {rfact} {ifact} {rfibc} {ifib} {lsearch} {bsearch} {bubble} {insertion}"
        output.append(row)
    return output
```

Q2. 1.5 Runtime and Complexity Comparison Table (5 marks)

```
from math import log2 , floor
def generate_runtime_complexity_table(n):
    # your code goes here
    output=[
        "Runtime Complexity Comparison",
        "Method ObservedCount ExpectedComplexity Observation",
        f"Linear Search {n} O(n) Grows linearly",
        f"Binary Search {floor(log2(n))+1} O(log n) Grows logarithmically",
        f"Bubble Sort {n*(n-1)//2} O(n^2) Grows quadratically",
        f"Insertion Sort {n*(n-1)//2} O(n^2) Grows quadratically"
    ]
    return output
```